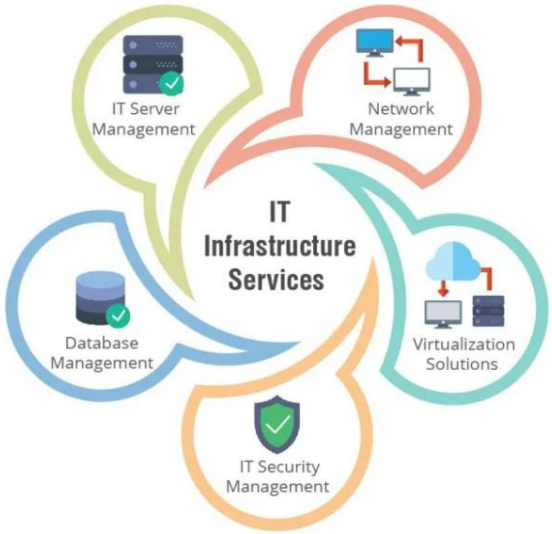
Introduction to System Administration & IT Infrastructure Services



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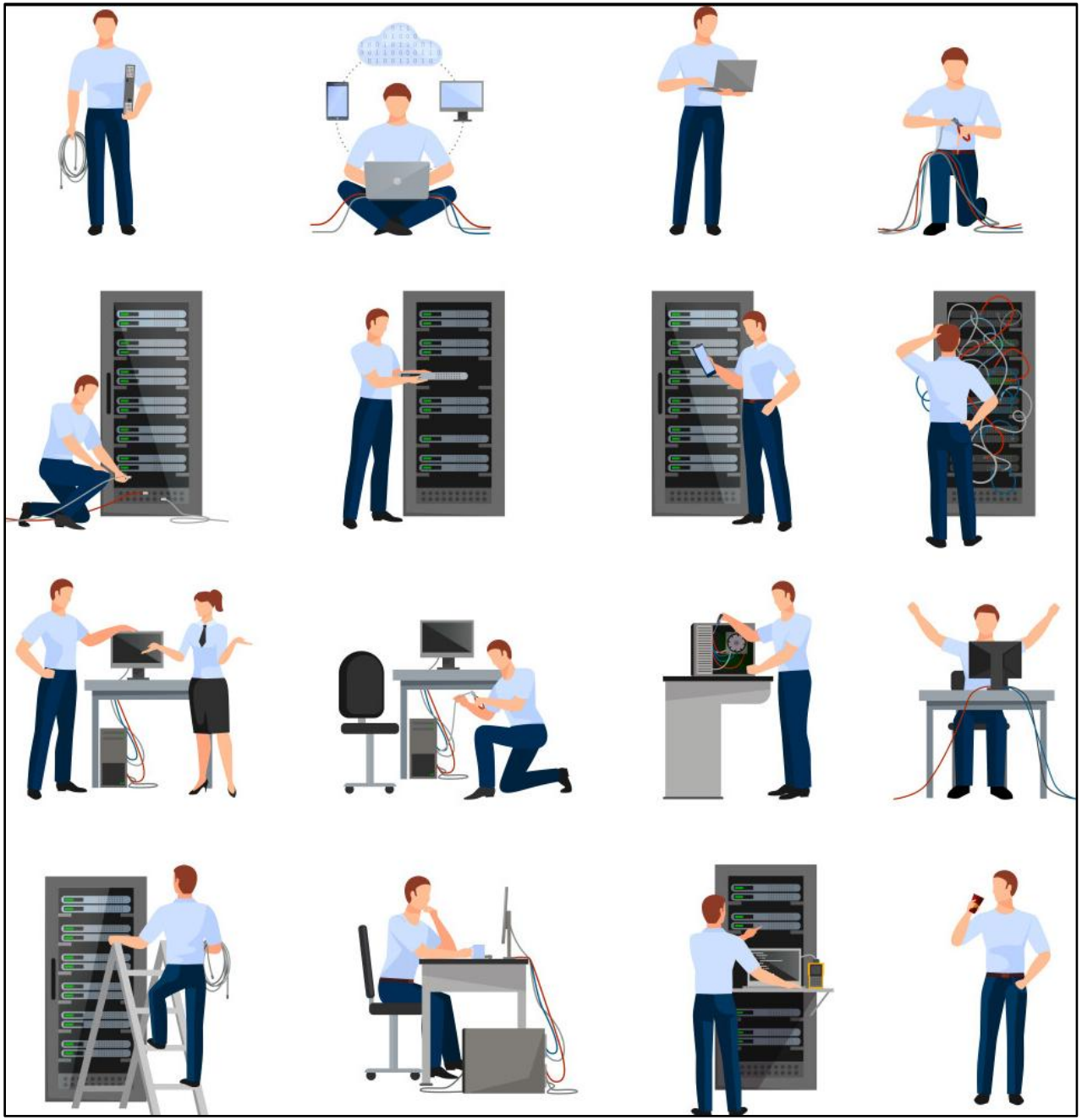
System Administration : Definition ??

System administration involves the management, maintenance, and operation of computer systems and networks within an organization.

- ✓ System administrators, often referred to as sysadmins, are responsible for ensuring that computer systems, servers, networks, and associated hardware and software function efficiently and securely to support the organization's goals and objectives.



Key Responsibilities : System Administrators

1. **Installation and Configuration**: Setting up and configuring hardware, operating systems, and software applications according to organizational requirements.
2. **Maintenance and Monitoring**: Regularly monitoring system performance, diagnosing and troubleshooting hardware and software issues, and performing routine maintenance tasks such as updates, backups, and security patches.
3. **Security Management**: Implementing and maintaining security measures to protect systems and data from unauthorized access, viruses, malware, and other cyber threats. This involves setting up firewalls, encryption, access controls, and conducting security audits.
4. **User Support**: Providing technical support and assistance to users, helping them with system-related issues, answering queries, and providing training when necessary.

Key Responsibilities : System Administrators

5. **Backup and Disaster Recovery**: Implementing backup and disaster recovery plans to ensure data integrity and minimize downtime in the event of system failures, natural disasters, or other emergencies.
6. **Documentation and Reporting**: Documenting system configurations, procedures, and troubleshooting steps for future reference, as well as preparing reports on system performance, security incidents, and compliance.
7. **Capacity Planning**: Analyzing system usage patterns and resource requirements to anticipate future needs and ensure that systems can scale effectively to accommodate growth.

THE ESSENTIAL ROLE OF A SYSTEM ADMINISTRATOR

Mastering Digital Infrastructure

DISASTER RECOVERY

Ensure data backups and recovery plans are in place.

SECURITY MANAGEMENT

Maintain security protocols and implement updates.

TECHNICAL SUPPORT

Provide technical support to users.

SYSTEM MANAGEMENT

Manage and maintain computer systems and networks.

SOFTWARE CONFIGURATION

Install and configure software and hardware.

PERFORMANCE MONITORING

Monitor system performance and troubleshoot issues.

The Human Capital Hub

QUALITIES OF AN ENTERPRISE SYSTEM ADMIN

FILECLOUD

Procedural Knowledge and Components

Must be Discipline

Need to Work With Non-Technical Staff

A Good Sense of Judgement

Reliable Strong Technical Knowledge

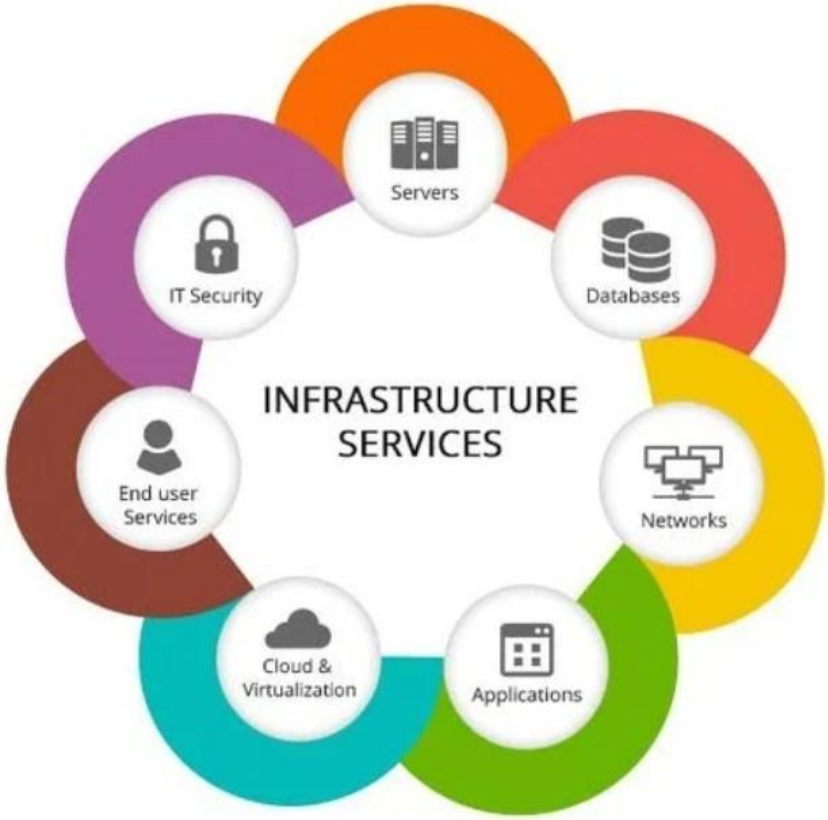
Communication Skills

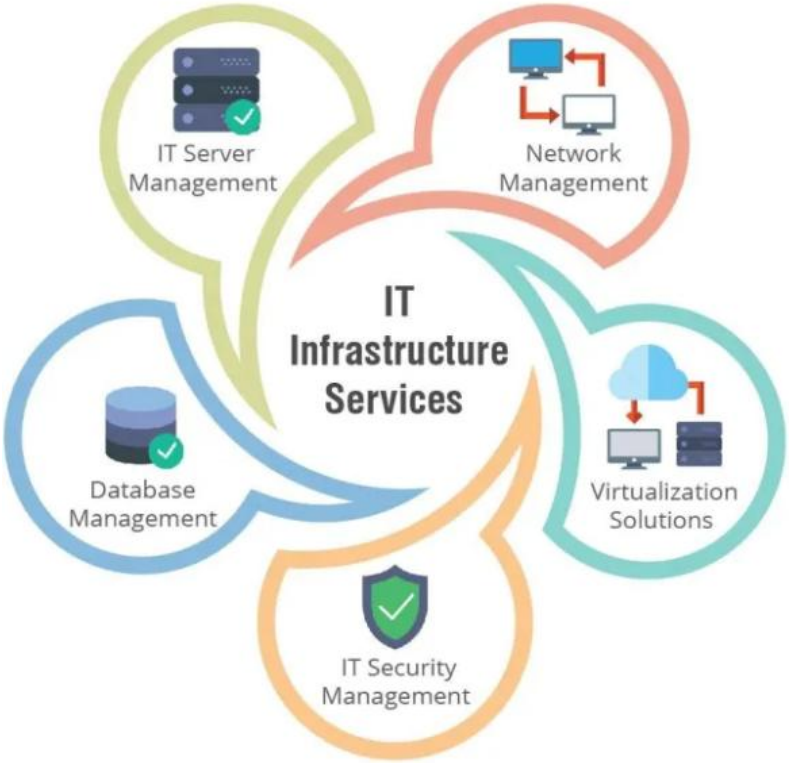
www.filecloud.com

IT Infrastructure Services : ??

- ✓ IT infrastructure services refer to a range of foundational IT components and capabilities that support the operation, management, and delivery of IT services within an organization.

- ✓ These services encompass hardware, software, networking, and other resources necessary to enable the functioning of an organization's IT environment.


INFRASTRUCTURE SERVICES
Servers
Databases
Networks
Applications
Cloud & Virtualization
End user Services
IT Security


IT Infrastructure Services
IT Server Management
Network Management
Virtualization Solutions
IT Security Management
Database Management

IT Infrastructure Services: Common Types ??

- **Hardware Services**: This includes the provision, maintenance, and management of physical hardware components such as servers, storage devices, network switches, routers, and desktop computers. Hardware services ensure that IT systems have the necessary computing power, storage capacity, and networking capabilities to support business operations.
- **Software Services**: This involves the deployment, configuration, and maintenance of software applications and systems used within the organization. Software services may include operating systems, productivity suites, database management systems, enterprise resource planning (ERP) software
- **Networking Services**: Networking services encompass the design, implementation, and management of network infrastructure, including local area networks (LANs), wide area networks (WANs), wireless networks, and internet connectivity.

IT Infrastructure Services: Common Types ??

- **Security Services**: Security services focus on protecting IT infrastructure, data, and systems from unauthorized access, cyber threats, and security breaches. This includes implementing firewalls, intrusion detection systems (IDS), antivirus software, encryption, access controls, and security policies to safeguard sensitive information and ensure compliance with regulatory requirements.
- **Storage and Backup Services**: Storage and backup services involve the management of data storage infrastructure, including storage area networks (SANs), network-attached storage (NAS), and cloud storage solutions.
- **Virtualization Services**: Virtualization services enable the creation and management of virtualized computing environments, allowing multiple virtual machines (VMs) to run on a single physical server. Virtualization improves resource utilization, scalability, and flexibility while reducing hardware costs and energy consumption.

IT Infrastructure Services: Common Types ??

- **Monitoring and Management Services**: Monitoring and management services involve the monitoring of IT infrastructure components and systems to ensure optimal performance, availability, and reliability. This includes monitoring system metrics, detecting and resolving issues proactively, and performing routine maintenance tasks such as software updates and patches.

- **Cloud Services**: Cloud services involve the provision of IT resources and services over the internet, including infrastructure as a service (IaaS), platform as a service (PaaS), and software as a service (SaaS). Cloud services provide scalability, flexibility, and cost-effectiveness by allowing organizations to access computing resources on-demand without the need for on-premises infrastructure.

Importance of System Administration : IT Infrastructure

- **Maintaining System Availability**: System administrators ensure that IT systems, servers, and networks are available and accessible to users when needed. This helps minimize downtime and ensures that employees can perform their tasks efficiently without disruption.
- **Ensuring System Reliability**: System administrators are responsible for ensuring the reliability and stability of IT systems and infrastructure. They monitor system performance, diagnose and troubleshoot issues, and implement preventive maintenance measures to prevent system failures and data loss.
- **Enhancing Security**: System administrators play a crucial role in implementing and maintaining security measures to protect IT systems and data from unauthorized access, cyber attacks, malware, and other threats. This includes configuring firewalls, implementing access controls, applying security patches, and conducting regular security audits.
- **Managing User Access**: System administrators are responsible for managing user accounts, permissions, and access rights to ensure that only authorized users have access to sensitive data and resources. They enforce security policies and procedures to protect against insider threats and unauthorized access.

Importance of System Administration : IT Infrastructure

- **Supporting Business Continuity**: System administrators develop and implement backup and disaster recovery plans to ensure that critical data and systems can be restored quickly in the event of system failures, natural disasters, or other emergencies. This helps minimize downtime and ensures business continuity.
- **Optimizing System Performance**: System administrators monitor system performance metrics and identify areas for optimization and improvement. They fine-tune system configurations, allocate resources effectively, and implement performance tuning measures to ensure optimal performance and scalability.
- **Managing IT Resources**: System administrators are responsible for managing IT resources such as hardware, software licenses, and networking equipment. They ensure that IT resources are utilized efficiently, cost-effectively, and in alignment with organizational goals and priorities.
- **Providing Technical Support**: System administrators provide technical support and assistance to users, helping them with system-related issues, troubleshooting problems, and providing guidance and training as needed. This improves user productivity and satisfaction with IT services.

History and Evolution of System Administration

Mainframe Era (1950s-1960s)

- During this period, large mainframe computers were the primary computing platforms used by organizations and institutions.
- System administration tasks were primarily performed by skilled operators who managed the hardware and software configurations of these mainframe systems.

Introduction of Minicomputers (1960s-1970s)

- The advent of minicomputers, smaller and more affordable than mainframes, led to the decentralization of computing power.
- System administration tasks expanded to include managing multiple minicomputers, each serving different departments or functions within organizations.

History and Evolution of System Administration

An IBM System/360.

History and Evolution of System Administration

History and Evolution of System Administration

Unix Revolution (1970s-1980s):

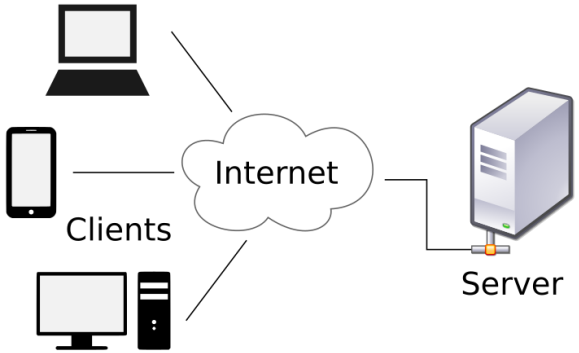
- The development of the Unix operating system by Bell Labs in the early 1970s marked a significant milestone in the evolution of system administration.
- Unix introduced concepts such as multi-user, multitasking environments, file system hierarchies, and shell scripting, which laid the foundation for modern system administration practices.
- System administrators became responsible for managing Unix-based systems, performing tasks such as user account management, system configuration, and software installation.

History and Evolution of System Administration

History and Evolution of System Administration

Client-Server Computing (1980s-1990s)

- The proliferation of personal computers (PCs) and the adoption of client-server computing architectures transformed the IT landscape.
- System administrators were tasked with managing networks of interconnected PCs and servers, ensuring seamless communication and data sharing among users.

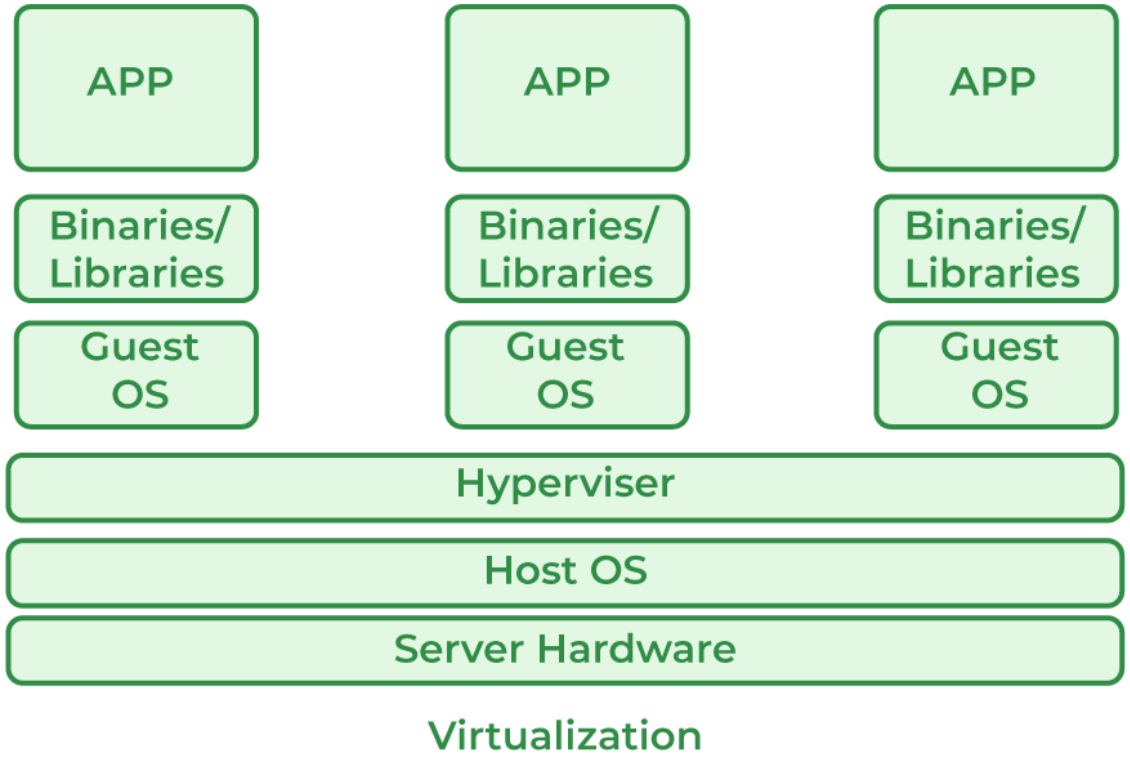


History and Evolution of System Administration

Internet Age (1990s-Present)

- The widespread adoption of the internet and the World Wide Web brought new challenges and opportunities for system administrators.
- System administrators played a crucial role in managing web servers, email systems, and other internet-facing services, ensuring their availability, security, and performance.
- The rise of e-commerce, cloud computing, and virtualization further expanded the scope of system administration, requiring administrators to adapt to new technologies and paradigms.

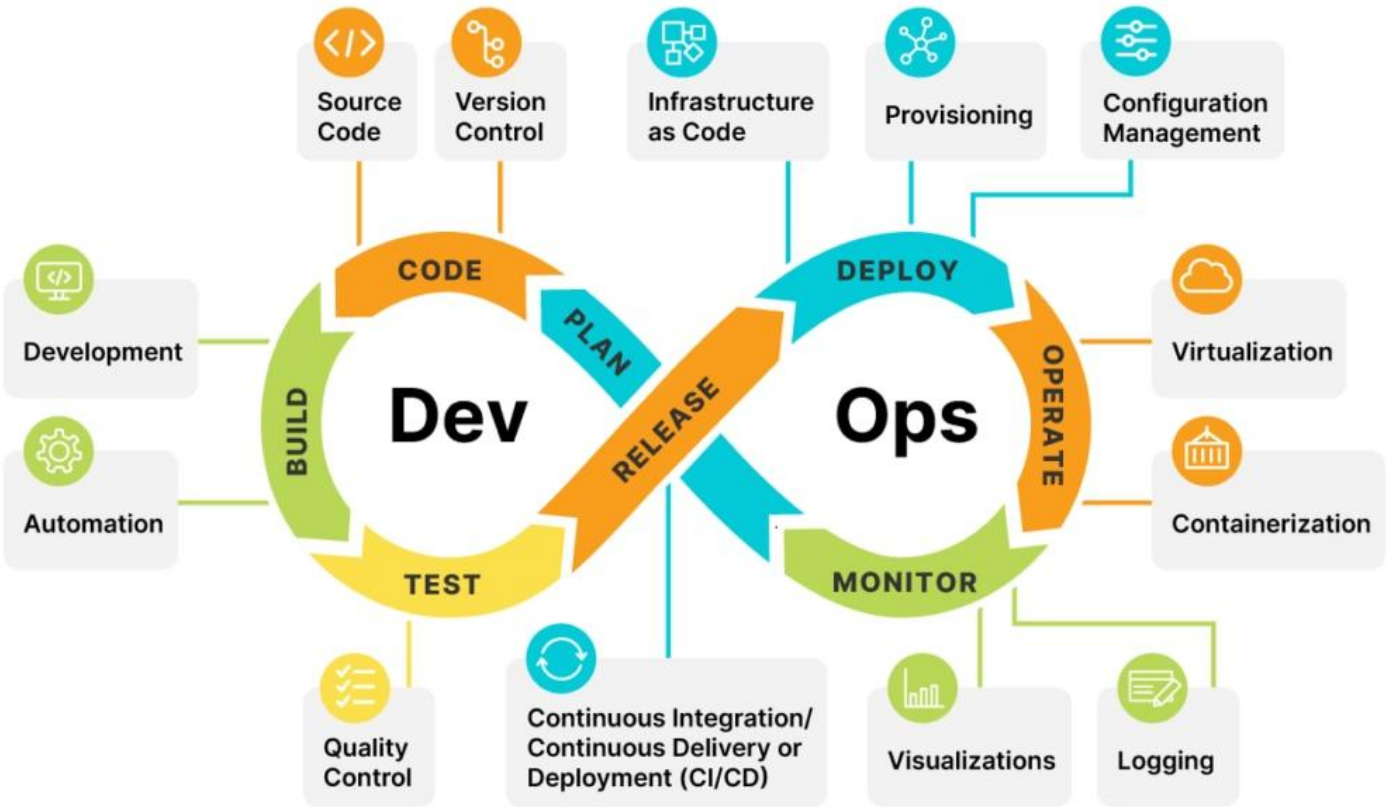
History and Evolution of System Administration



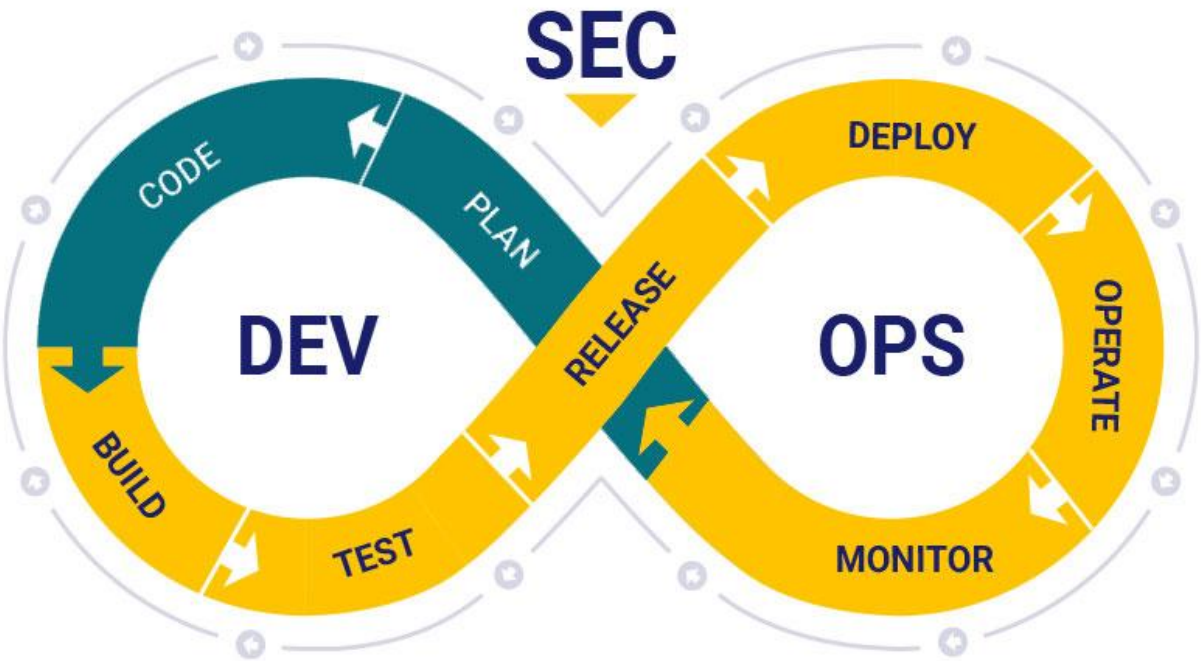
History and Evolution of System Administration

- **DevOps and Automation (2000s-Present)**:
  - The emergence of DevOps practices, emphasizing collaboration between development and operations teams, revolutionized the way IT systems are managed and deployed.
  - Automation tools and technologies such as configuration management, Infrastructure as Code (IaC), and continuous integration/continuous deployment (CI/CD) pipelines have become essential for modern system administration.
  - System administrators increasingly focus on automating routine tasks, streamlining processes, and adopting agile methodologies to meet the demands of rapidly evolving IT environments.

History and Evolution of System Administration



History and Evolution of System Administration



Components of IT infrastructure

Hardware

The physical component of an IT infrastructure. It comprises elements such as computer or storage devices.

Software

The applications that allow the hardware to perform critical tasks. Its common example is an operating system.

Network

The core of an IT infrastructure, it requires both hardware and software to function and facilitates connectivity and resource sharing.

Data center

The physical or virtual space to store and manage business information and applications.

Capterra

5 Types of IT Services

1 Managed services

- Outsourcing daily IT management
- Remote monitoring, network admin, cybersecurity
- Frees up in-house IT for strategic work

2 Cloud computing services

- Store and access data over the internet
- Main types: SaaS, PaaS, IaaS
- Google Workspace, Salesforce's Heroku, AWS

3 Network services

- Setup, management, and security of network systems
- Ensures device connectivity
- Vital for preventing failures and breaches

4 Software development and implementation

- Tailored software solutions for business needs
- Integration into existing systems
- Customized IT solutions

5 Support and help desk services

- Troubleshoots IT-related issues
- Addresses minor glitches to major outages
- Offers ongoing support within an organization

Techopedia

Service Catalog
Service Desk
Incident Management
Problem Management
Change Management
Asset Management
Knowledge, Policy and Procedure
Service Strategy
Reports & Dashboard
Service Improvement
ITSM

IT Service Management : ITSM

- IT Service Management (ITSM) refers to the set of practices, processes, and frameworks used by organizations to design, deliver, manage, and improve IT services to meet the needs of their customers and support their business objectives.

- ITSM focuses on aligning IT services with the needs of the business, optimizing service delivery, and ensuring the efficient use of IT resources.

IT Service Management : ITSM

- IT Service Management is typically supported by frameworks and standards such as ITIL (Information Technology Infrastructure Library), which provides a comprehensive set of best practices for ITSM processes, roles, and functions.

- Other frameworks and methodologies, such as COBIT (Control Objectives for Information and Related Technologies) and ISO/IEC 20000 (International Standard for IT Service Management), also provide guidance and certification criteria for implementing effective ITSM practices.

SLA: Service Level Agreement

- SLA stands for Service Level Agreement. It is a contractual agreement between a service provider and a customer that defines the level of service expected from the provider.

- SLAs are commonly used in various industries, including IT, telecommunications, and outsourcing, to formalize the relationship between service providers and their customers and to establish clear expectations regarding service quality, performance, and availability.

Key Components of SLA

1. **Service Description**: A description of the services provided by the service provider, including details such as service scope, features, and functionalities.

2. **Service Level Objectives (SLOs)**: Specific, measurable targets or objectives that the service provider agrees to meet regarding service performance, availability, and quality. SLOs are often defined in terms of metrics such as uptime, response time, and resolution time.

3. **Responsibilities and Obligations**: The roles and responsibilities of both the service provider and the customer, including any requirements for cooperation, communication, or assistance in achieving the agreed-upon service levels.

Key Components of SLA

4. **Performance Measurement and Reporting**: Procedures for measuring, monitoring, and reporting on the performance of the services provided, including the frequency and format of performance reports and the mechanisms for resolving disputes or discrepancies.

5. **Penalties and Remedies**: Provisions for penalties or remedies in the event that the service provider fails to meet the agreed-upon service levels. This may include financial penalties, service credits, or other forms of compensation for the customer.

6. **Termination and Renewal**: Conditions for terminating or renewing the SLA, including the notice period, termination clauses, and procedures for renegotiating or updating the agreement.

Thank You

Should you have any queries, please contact me at
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